

Quick Check

Lesson 4: Tangent and the Sine-Cosine-Tangent Relationship: Solving the Mango-Tree Problem

5 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.1: Trigonometry I | Lesson 4

In a right-angled triangle, which ratio equals $\tan \theta$?

A Opposite / Hypotenuse

B Adjacent / Hypotenuse

C Opposite / Adjacent

D Hypotenuse / Opposite

$\sin 60^\circ = 0.8660$ and $\cos 60^\circ = 0.5000$. Use $\tan \theta = \sin \theta / \cos \theta$ to find $\tan 60^\circ$.

A 0.5774

B 1.7320

C 0.4330

D 1.3660

The sun makes a 53° angle with the ground and a tree casts a 9.0 m shadow. Relative to the 53° angle, the shadow is the triangle's ...

- A Opposite side
- B Adjacent side
- C Hypotenuse
- D It depends on how tall the tree is

$\tan 53^\circ = h / 9.0$. Which step correctly finds h ?

A $h = 9.0 \div \tan 53^\circ$

B $h = 9.0 \times \tan 53^\circ$

C $h = \tan 53^\circ \div 9.0$

D $h = 9.0 + \tan 53^\circ$

A flagpole casts a 10 m shadow when the sun angle is 40° . $\tan 40^\circ = 0.8391$. How tall is the flagpole?

A 0.84 m

B 8.39 m

C 10.84 m

D 11.92 m

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).